

Drive-by-wire chassis — instruction manual

(English translation)

English translation of the Chinese instruction manual for Glorb's drive platform — the "**4WD 500M Remote Control Transport 72V**" electric four-wheel drive-by-wire chassis (see [../logistics/stuff.md](https://logistics/stuff.md)). Original file: [manual-zh-original.docx](#) · rendered PDF of this translation: [manual-en.pdf](#).

Manufacturer: **Fengxian Zhonglian Electric Technology Co., Ltd.** (丰县众联电动科技有限公司), D3 Steel City, Beiguan, Feng County, Xuzhou, Jiangsu, China. Contact: Manager Bu (卜经理), +86 153 3513 2286.

Translation notes. Several pages of the original are low-resolution scans; numbers that were hard to read are marked (?). Section numbering follows the original (it starts at "I. Function introduction" — the warranty card and troubleshooting table come before it, unnumbered). Original scanned tables are kept in [img/](#) alongside the translations.

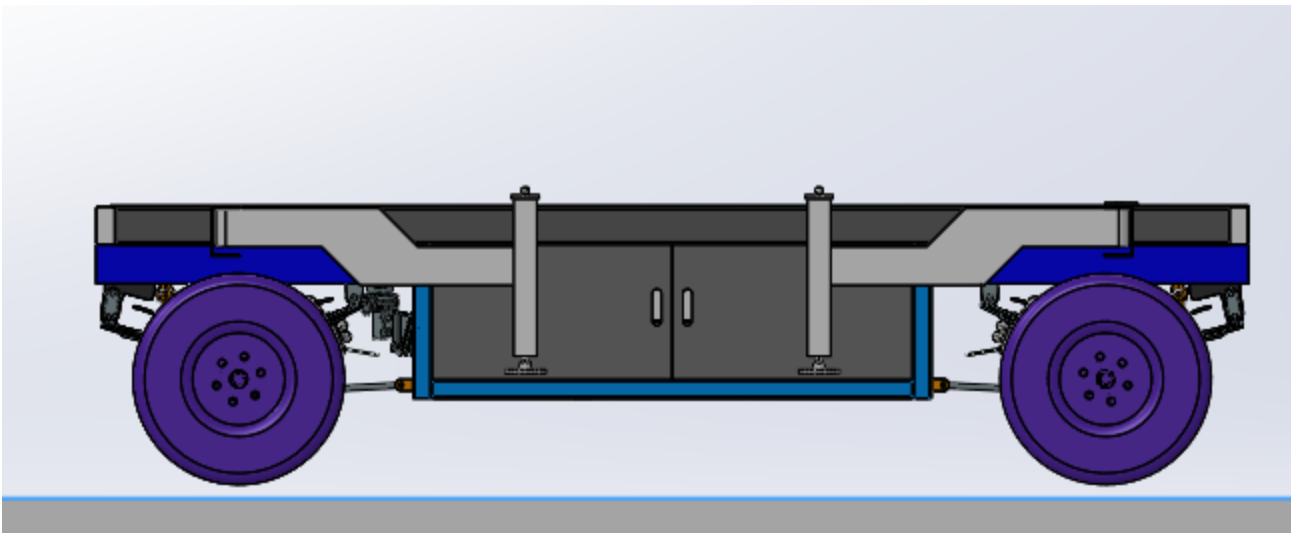
Quick pointers for Glorb debugging:

- The **beeping fault codes** from the motor controllers are in § IV — count the long and short beeps.
- Per § III and the wiring diagram, the **VCU (the control box the drone-style receiver plugs into), EPS steering, and EBS brakes all run on 12 V**, fed from the 60 V traction bus through DC-DC converters (50 A and 20 A). Each drive controller's key-switch wire (purple) and brake-signal wire (blue) get 60 V+ from the VCU.
- Remote pairing/troubleshooting is in the [troubleshooting table](#) and [remote operation](#).

Cover

INSTRUCTION MANUAL (使用说明书) Electric four-wheel drive-by-wire chassis (电动四轮线控底盘)

(Cover art: "Explore · Pioneer" — preserved in the original docx.)



Warranty information

User information: product model · purchase date · customer name.

Warranty record: warranty date · cause of fault and resolution · handler's signature · customer confirmation.

Warranty terms

- When purchasing this product, fill in this card carefully and read the warranty terms below to ensure effective warranty coverage:
 - Keep this card safe when purchasing, and have the seller stamp it to confirm.
 - This warranty card must be presented when requesting warranty service.
 - The information on this card must be truthful, otherwise it is void.
 - The warranty period is **one year**. If the product fails during the warranty period due to defective components or manufacturing problems, the company provides free repair and parts replacement.
- Damage from the following causes is **not** covered by the warranty:
 - Damage caused by installation or use not in accordance with this manual.
 - Any man-made or accidental damage.
 - Repairs or modifications not approved by the company, or a broken warranty seal sticker.
 - Aging, dents, or scratches of the product's outer housing.

Troubleshooting

Symptom	Check	Remedy
Steering / brakes have no power	Is the power connected incorrectly?	Check that the power-wire input and output ends are correct
	Are the ports loose, or has a power wire fallen off?	Reconnect the power wire and tighten the terminal screws

Symptom	Check	Remedy
Remote range is short, or remote does not work	Is it being operated next to strong magnetic fields or high-power electrical equipment? Are there too many obstructions between the control box and the transmitter?	Install the controller at least 0.5 m from high-power / strongly magnetic equipment; clear obstructions between the control box and the transmitter
	Was a CAN wire or power wire accidentally knocked loose?	Power-cycle the remote and the control box, and check the power supply
Buttons feel obstructed when pressed	—	Check the gap between the button and the panel for trapped debris, and clear it promptly
Remote transmitter cannot be used at all	Not yet paired with the control box	Re-pair the transmitter with the control box (see the pairing instructions on page 2 — the remote control operation section)
	Buttons fallen off or damaged	Replace or send for repair

I. Function introduction

The four-wheel-drive, dual-steering drive-by-wire vehicle has high obstacle-crossing capability, with all four wheels driven simultaneously. It receives signals over the **CAN bus** to control forward / reverse / steering / braking. The product can be paired with multiple wireless control panels for flexible multi-controller operation. The body uses a compact design and is easy to operate.

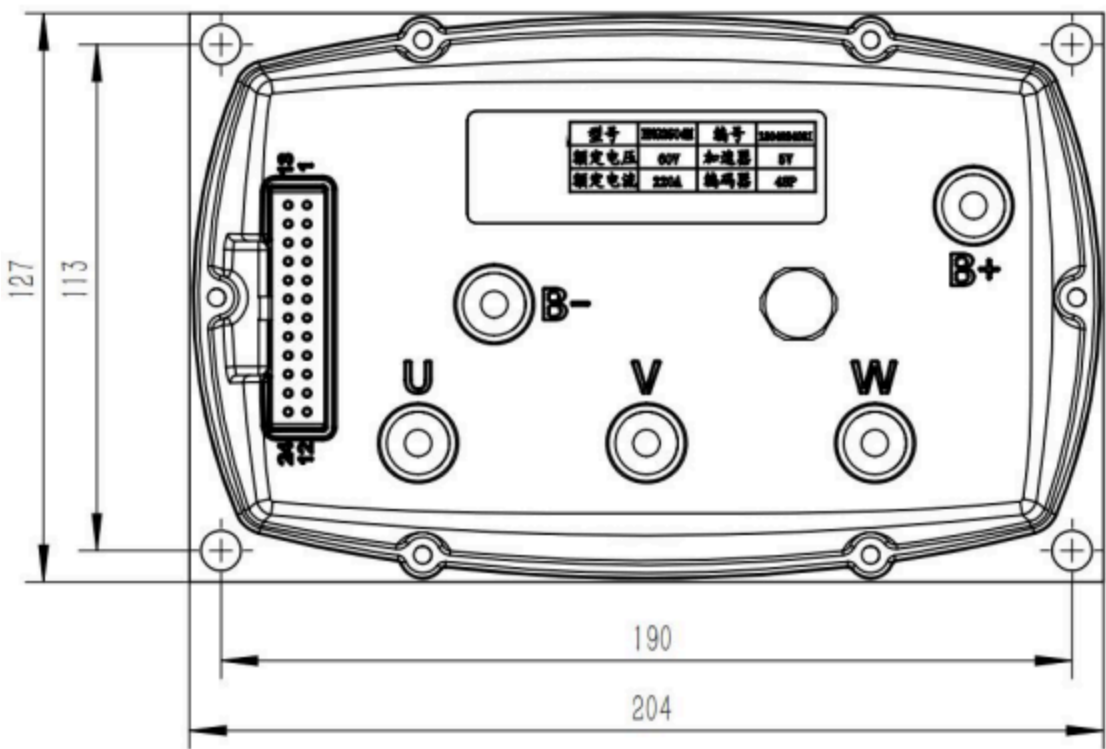
II. Product components / technical specifications

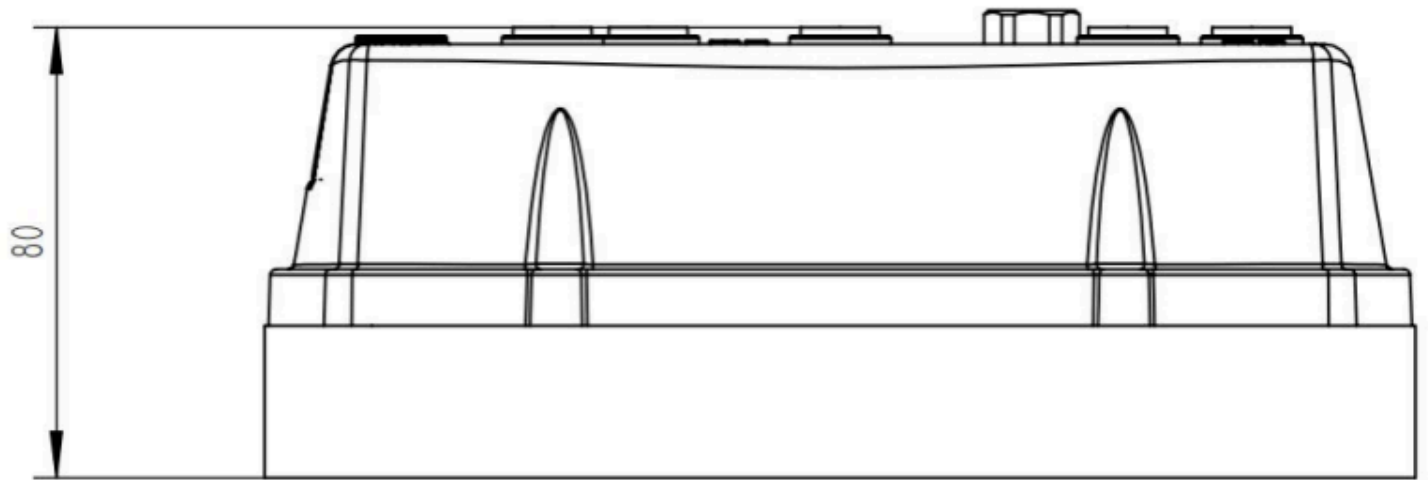
AC drive controller



Nameplate (from the dimension drawing): rated voltage **60 V**, rated current **220 A**, throttle input **5 V**, encoder **48 P**.

Dimensions: 204 × 127 × 80 mm overall; mounting holes 190 × 113 mm.





Notes from the drawing:

1. **B+** connects to battery positive, **B–** to battery negative; **U, V, W** connect to motor U, V, W respectively.
2. U/V/W motor cables and battery cables: 20 mm² cross-section.
3. Terminal posts are secured with M6×20 bolts, torqued to 6–10 N·m.
4. The motor controller complies with GB/T 18488.1-2015 and GB/T 18488.2-2015 (Chinese EV drive-motor standards).
5. Untoleranced dimensions per GB/T 180-M.
6. All terminal directions shown in the drawing are the wire-entry direction.

CAN protocol (data format)

Original scan: [img/can-protocol-zh.png](#)

Frame 1 — ID 0x1A0 : vehicle controller (TCU) → motor drive (MCU), speed control frame, period 100 ms

Byte	Field	Definition
1	Motor enable state	0x00 : disabled · 0xA5 : enabled
2	Switch (gear) info	0x00 : neutral · 0xAA : forward · 0x55 : reverse · 0x88 : brake
3-4	Target speed	0–6000 rpm, high byte last
5-6	—	—
7-8	0x7889	Message identification, high byte last

Frame 2 — ID 0x1B0 : MCU → all, feedback frame, period 100 ms

Byte	Field	Definition
1-2	Motor speed	offset –10000, 1 rpm/bit, high byte last
3-4	Motor current	1 A/bit
5	Alarm code	see § IV
6	Controller temperature	1 °C/bit, offset –40

Byte	Field	Definition
7	Motor temperature	1 °C/bit, offset –40 (<i>scan reads "–41"</i>)
8	Driving-state flag	0x00 : manual (RC) control · 0xA5 : CAN control

Frame 3 — ID 0x00 : reserved — do not use this ID for anything else.

AC motor with electromagnetic brake



Motor model **JS-YS140-H22** — rated DC 60 V, 2.2 kW, 4 poles, insulation class H, protection rating IP55. Output spline: 18 teeth, module 1, pressure angle 30°, major Ø19.5 mm, minor Ø17.11 mm. ([Original parameter scan](#), [dimension drawing](#).)

Dynamometer characteristic points (*low-quality scan — treat digits as approximate*):

Point	V	A	Input kW	Torque N·m	rpm	Output kW	Eff. %	Time s
No load	62.7	32.3	2.03	0	4514	0	0	0
Rated	62.3	38.3	2.39	4.74	4475	2.22	93.1	8
Max efficiency	62.3	38.3	2.39	4.74	4475	2.22	93.1	6
Max output	61.4	198.8	12.21	48.42	1418	7.19	58.9	64
Max torque	62.6	155.7	9.75 (?)	55.66	<402	2.34	24	76
End	62.6	155.7	9.75 (?)	55.66	0	0	0	76

Electrical connectors ([original scan](#)):

- **Black 4-pin connector — speed sensor** (HMEX67Q003 (?)), plug DJ7041-1.5: pin 1 V+ (red) · pin 2 signal B (green) · pin 3 signal A (white) · pin 4 ground (black). The sensor outputs **48 pulses per revolution**.

- **Black 2-pin connector — temperature feedback**, plug DJ7021-1.5: pin 1 "+" (red) · pin 2 "—" (black). Sensor: **KTY84-150**.

EPS (electric power steering)



Parameter	Value
Product	EPS assembly, steering-column type
Working current	65 A max
Motor torque	3.4 N·m
Reduction ratio	1 : 17
Working temperature	−40 ... 105 °C
Working voltage	12 V nominal (9–16 V)
Relative humidity	93 % (500 h) @ 40 °C
Spline-to-spline distance	178 mm
Input spline	40 teeth (module 0.4722, Ø19.3)
Output spline	36 teeth (module 0.47, Ø17.3)

Drawings: [EPS dimensions](#) · [EPS ECU photo](#) · [ADAS EPS-ECU wiring & protocol sheet](#) (scan too low-resolution to translate in full; it is the ECU's connector/pin sheet).

EBS (electronic brake booster)



Parameter	Value
Product	Electronic brake booster assembly, model EBS-15T (A)
Weight	7.5 kg
Mounting envelope	345 × 253 × 197 mm
Application	vehicles < 1.5 t (speeds ≥ 60 km/h (?)), max boost 5.2 kN (?); vehicles ≤ 4 t at ≤ 40 km/h if matched to caliper displacement, max boost 5.2 kN (?)
Master cylinder	plunger-type tandem dual-chamber, ports ISO M10×1.0, bore Ø20.64 mm, effective stroke 19 + 19 mm
Pedal rod	optional with or without, per customer requirement
Working temperature	−40 ... 105 °C
Working voltage	12 V nominal (10–14 V)
Motor rated power	200 W typical, 500 W max (?)
Bus current	20 A typical; stall current higher (<i>digits unreadable</i>)
Max output pressure	10–14 MPa (?)
Pressure build time (10→90 %)	150 ms
Pressure release time (90→10 %)	100 ms

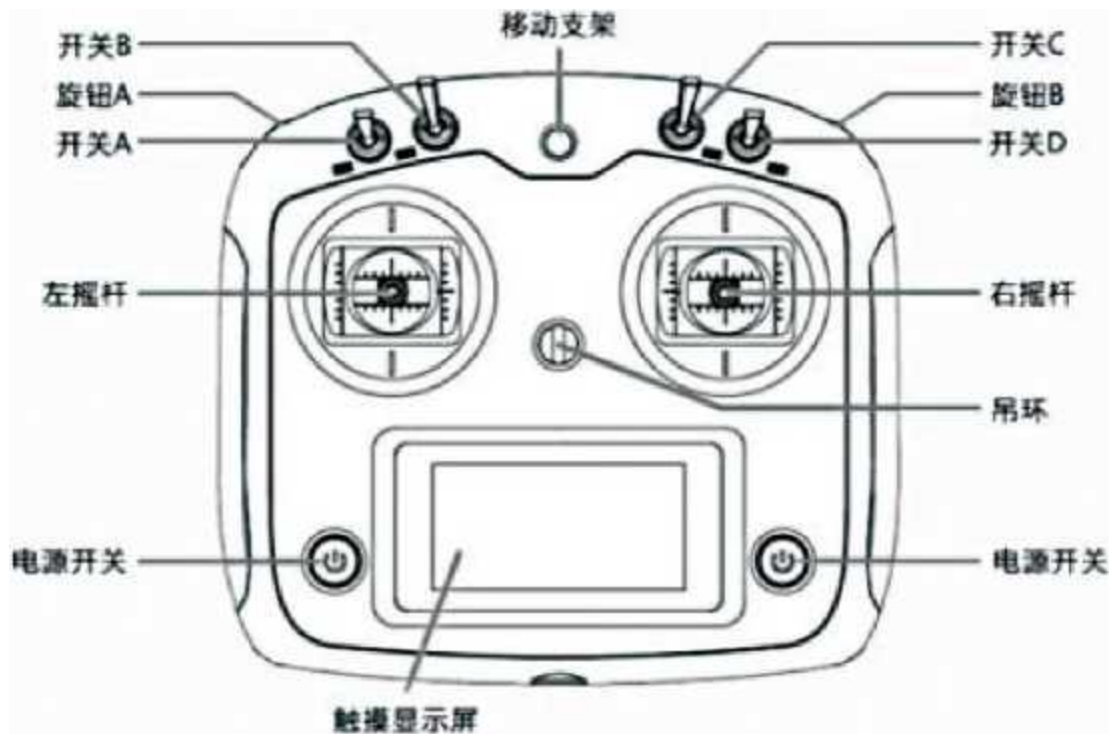
Drawings: [front](#) · [side](#) · [top](#)

EBS alarm codes (bit field)

Original scan: [img/ebs-alarm-bits-zh.png](#)

Bit	Fault	Code	Notes
0	Undervoltage	01	transient voltage too low
1	Overload	02	
2	Overvoltage	04	transient voltage too high
3	U-phase fault	08	
4	V-phase fault	10	
5	W-phase fault	20	
6	Overcurrent	40	
7	Stall protection	80	
8	IPM fault	01	drive fault
9	CAN timeout	02	communication timeout
10	Self-learning fault	04	magnetic-pole learning fault
11	12 V supply fault	08	sustained abnormal supply voltage (over- or undervoltage)
12	Self-check fault	10	sensor / MCU RAM / ROM self-check (internal program check)
13	busoff	20	reserved
14	Reserved	40	
15	Ignition-signal fault	80	ignition signal disconnected

Remote control operation



Layout diagram labels (clockwise from top): phone/mobile mount (top center), switch B and switch C (inner toggles), switch A and switch D (outer toggles), knobs A and B, left stick, right stick, lanyard ring (center), touch display (bottom center), and **two power switches** (bottom left + bottom right).

Power-on:

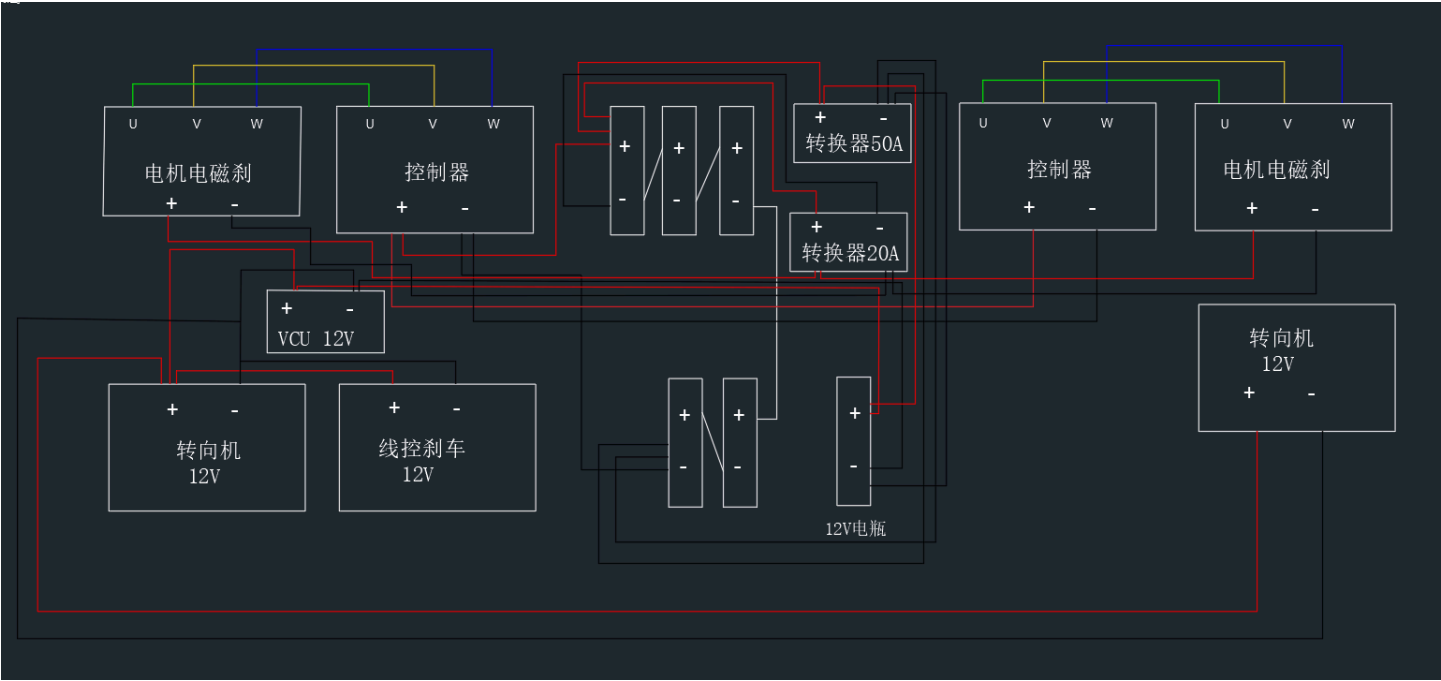
1. Press **both power switches at the same time** to turn the transmitter on.

2. Flip **all toggle switches to the up position**; when the display shows "flight mode off" (飞行模式关闭), the remote is ready to use.

Switch functions:

Switch	Position	Function
A	up	normal remote-control mode
A	down	host-computer CAN control mode (drive commands come from an upstream computer instead of the sticks)
B	up	forward
B	middle	neutral
B	down	reverse
C	up	turn on switched output 1
C	middle	turn off switched outputs 1 and 2
C	down	turn on switched output 2
D	up	turn off switched output 3
D	down	turn on switched output 3

III. Wiring and pairing



Wiring-diagram blocks (translated): 电机电磁刹 = motor with electromagnetic brake (×2, U/V/W phases) · 控制器 = drive controller (×2) · VCU 12V = vehicle control unit · 转向机 12V = steering motor / EPS (×2) · 线控制车 12V = brake-by-wire / EBS · 转换器50A / 转换器20A = 50 A and 20 A DC-DC converters · center columns = traction battery bank · 12V电瓶 = 12 V battery.

1. + connects to battery positive, – to battery negative; controller **U, V, W** connect to motor **V, U, W** respectively (sic — the wiring note swaps U and V here, while the controller drawing says U→U, V→V, W→W; follow whatever the vehicle is actually wired as).

2. The steering **EPS**, brake **EBS**, and **VCU** each take one set of 12 V positive and negative from the **12 V supply**; the EPS and EBS **ignition wires** also connect to 12 V+.
3. All **CAN H** lines of the EPS, EBS, VCU, and drive controllers connect together to CAN H, and all **CAN L** lines to CAN L.
4. All drive-controller positives connect to **60 V+**, all negatives to **60 V-**.
5. Each drive controller's **key-switch wire (purple)** and **brake-signal wire (blue)** connect to **60 V+ on the body controller** (the VCU control box).

(Pairing the transmitter to the control box: see the power-on procedure in [remote control operation](#); the troubleshooting table calls this "pairing". The transmitter/receiver pair is a FlySky-style 2.4 GHz RC system — if a full re-bind is ever needed, use the standard bind procedure for the transmitter model on the receiver inside the control box.)

IV. Drive controller fault codes

The drive controller signals faults as **audible beep patterns** ("one long, two short" = 1 long beep followed by 2 short beeps, repeating). Original scan: <img/fault-codes-zh.png>

Code	Beeps	Fault	Cause / analysis
#0011	1 long, 1 short	Overvoltage protection	battery voltage does not match controller
#0012	1 long, 2 short	Undervoltage protection	battery voltage does not match controller
#0013	1 long, 3 short	Capacitor-board low voltage	battery voltage low, or controller fault
#0014	1 long, 4 short	Power-module short circuit	controller fault
#0015	1 long, 5 short	Chip fault	controller fault
#0016	1 long, 6 short	Current sensor 1 fault	controller fault
#0017	1 long, 7 short	Current sensor 2 fault / current calibration / customization fault	controller fault
#0019	1 long, 9 short	CAN communication fault	—
#0021	2 long, 1 short	Throttle fault	throttle reading high / mismatched, or throttle needs reset
#0022	2 long, 2 short	Gear protection	reset gear selection to neutral
#0023	2 long, 3 short	Simultaneous throttle and brake request	brake signal mismatched or wrong signal threshold
#0024	2 long, 4 short	Charging protection alarm	disconnect the charging plug
#0025	2 long, 5 short	Simultaneous forward and reverse signals	gear wiring error

Code	Beeps	Fault	Cause / analysis
#0026	2 long, 6 short	Encoder fault	—
#0027	2 long, 7 short	Encoder signal lost	—
#0031	3 long, 1 short	Circuit supply	key-switch poor contact
#0032	3 long, 2 short	Phase-wire overcurrent protection	three-phase wiring poor contact, encoder unstable, or motor phase short
#0033	3 long, 3 short	Controller over-temperature protection	let it cool, move to a better-ventilated location, or add heatsinking
#0034	3 long, 4 short	Motor over-temperature protection	let it cool, or fit a higher-power motor
#0035	3 long, 5 short	Temperature-sensor fault or low-temperature protection	stops working below $-25\text{ }^{\circ}\text{C}$
#0051	—	Overcurrent protection	phase-wire or supply short circuit
#0052	—	Overcurrent protection	phase-wire or supply short circuit
#0053	—	Overcurrent protection	phase-wire or supply short circuit

V. Safety notes

- Install and operate this product strictly according to this manual. Operators must have basic electrical knowledge, and should work **with power disconnected** to avoid electric shock.
- The vehicle's electronic control components must be installed in a dry environment free of flammable or explosive gases. Never use this product in damp conditions or scenarios without any protective measures.
- The product's applicable supply voltage is AC 48 V-220 V (*sic — this line appears to refer to the charger input*); connecting a supply outside this range will damage the product.
- If the housing is damaged, or the product emits abnormal noises or smells during use, stop using it immediately and repair or replace it.

Back cover: PRODUCT SPECIFICATION (产品说明书) — "Providing excellent service is our mission; what we bring customers is never just an excellent product." Fengxian Zhonglian Electric Technology Co., Ltd., D3 Steel City, Beiguan, Feng County, Xuzhou, Jiangsu · Manager Bu, +86 153 3513 2286.